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Main tilting door for closing an access opening of a room

Abstract

A main tilting door for the closure of an access opening of a room includes two fixed uprights, including vertical sliding guides, and one fixed crosspiece associable with each other to define a fixed frame. The door includes two movable uprights and two movable crosspieces associable with each other to define a movable frame. The door includes tilting movement device/unit/component/apparatus or the like including two lateral supporting arms each having a first end which can be hinged to an intermediate area of the movable uprights and a second end which can be hinged to a top of the fixed uprights; and two sliding supports movable along the vertical sliding guides. The door includes panels associable with the movable frame. At least one of the movable uprights and the movable crosspieces includes a sliding track and cursors movable along the sliding track, the panels associable with the cursors.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2150321	12/1938	Ferris	49/205	E05D 15/445
3038568	12/1961	Morgan	52/235	E04B 2/90
3407553	12/1967	Halpern	49/505	B66B 13/306
3695332	12/1971	Bahnsen	292/225	E05F 15/686
4653565	12/1986	Iha	160/193	E05F 15/686
4819376	12/1988	Taddei	N/A	N/A
5038538	12/1990	Rozon	52/509	E06B 1/526
5365993	12/1993	Jella	160/232	E05F 15/673
5373663	12/1993	Turini	49/200	E05F 15/681
5444958	12/1994	Lu	52/771	E06B 3/5454
5488982	12/1995	Rejc	160/310	E06B 9/15
5682719	12/1996	Huang	52/775	A47G 5/00
2007/0068097	12/2006	MIerAU	N/A	N/A
2010/0077664	12/2009	Stensland	160/181	E06B 3/721
2015/0376933	12/2014	Schweiss	52/698	E05F 15/622
2016/0032640	12/2015	Wang Chen	49/502	E06B 3/5892

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0 264 876	12/1987	EP	N/A
2978788	12/2012	FR	N/A
1 045 351	12/1965	GB	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to IT patent application No. 102019000000893 filed on Jan. 21, 2019, and this application claims priority to and is a 371 of international PCT Application No. PCT/IB2020/050375 filed on Jan. 17, 2020, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to a main tilting door for the closure of an access opening of a room, particularly of a garage space.

BACKGROUND ART

(3) Main tilting doors are closure systems in which an overhead door is moved with respect to a fixed frame by means of a system of articulation arms, hinges and counterweights.

(4) The overhead door has height and width equal to the entire access opening to be closed.

(5) In the closure position, in fact, the overhead door is arranged vertically and occupies the access opening area of the garage space, closing it.

(6) During opening, on the other hand, the overhead door tends to be placed horizontally, at the top of the fixed frame, freeing the passage through the access opening.

(7) Main tilting doors of known type do have several drawbacks, including the fact that they consist of a plurality of components that are difficult to assemble and mount.

(8) Therefore, the construction of traditional main tilting doors is carried out in the factory, using special equipment in assembly lines, with the consequence that the main tilting door thus assembled has very big size and weight and can only be transported to the construction site with considerable difficulty.

(9) The installation on site of this type of main tilting doors is also difficult to carry out, as it requires the use of special lifting equipment (cranes) and qualified personnel for installation, with a consequent increase in costs.

(10) As a result, the supply and assembly are carried out exclusively by specialized companies, thus limiting the spread of such main tilting doors and making them unattractive for customers.

Description of the Invention

(11) The main aim of the present invention is to devise a main tilting door for the closure of an access opening of a room, particularly of a garage space, which allows simple and practical assembly.

(12) A further object of the present invention is to obtain a main tilting door the components of which are of reduced size and weight, so that they can be easily moved and transported.

(13) Another object of the present invention is to enable assembly on site that is free of the need for bulky and expensive transport and lifting equipment and highly skilled labor.

(14) Another object of the present invention is to devise a main tilting door for the closure of an access opening of a room, particularly of a garage space, which allows overcoming the

aforementioned drawbacks of the prior art in the ambit of a simple, rational, easy, effective to use and low cost solution.

(15) The objects set out above are achieved by the present main tilting door for the closure of an access opening of a room, particularly of a garage space according to one or more aspects of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other characteristics and advantages of the present invention will be more evident from the description of a preferred, but not exclusive, embodiment of a main tilting door for the closure of an access opening of a room, particularly of a garage space, and of a method for the assembly on site of a main tilting door, illustrated as an indication, but not limited to, in the attached tables of drawings in which:

(2) FIG. 1 is an axonometric view of the main tilting door according to the invention;

(3) FIG. 2 is an exploded view of the main tilting door according to the invention;

(4) FIG. 2a is an exploded view of a detail of the main tilting door according to the invention;

(5) FIG. 2b is an exploded view of another detail of the main tilting door according to the invention;

(6) FIG. 3 is an axonometric view of the assembly phase of the fixed frame provided by the method according to the invention;

(7) FIG. 4 is an axonometric view of the assembly phase of the movable frame provided by the method according to the invention;

(8) FIG. 5 is an axonometric view of the hinging phase of the lateral supporting arms to the fixed uprights provided by the method according to the invention;

(9) FIG. 6 is an axonometric view of the association phase of the sliding supports with the movable uprights provided by the method according to the invention;

(10) FIG. 7 is an axonometric view of the association phase of the panels with the movable frame provided by the method according to the invention.

EMBODIMENTS OF THE INVENTION

(11) With particular reference to these figures, reference numeral 1 globally indicates a main tilting door for the closure of an access opening of a room, particularly of a garage space.

(12) The main tilting door 1 comprises at least two fixed uprights 2 and a fixed crosspiece 3 associable with each other to define a fixed frame 4 which can be fixed to an access opening of a room.

(13) For this purpose, the fixed uprights 2 comprise at least one coupling portion 5 which can be coupled by interlocking to a respective coupling seat 6 arranged on the fixed crosspiece 3.

(14) In particular, the coupling portion 5 consists e.g. of a first shaping that is obtained at the top of the fixed uprights 2 and that can be prismatically coupled to a second shape.

(15) The second shape is obtained at the opposite ends of the fixed crosspiece 3 and defines the coupling seats 6.

(16) After the coupling portions 5 have been coupled by interlocking to the coupling seats 6, the fixed uprights 2 and the fixed crosspiece 3 can be locked in place by means of screws and/or bolts.

(17) This ensures that the fixed uprights 2 and the fixed crosspiece 3 are assembled correctly and extremely easily and quickly, even during the installation of the main tilting door 1.

(18) The fixed uprights 2 are provided with vertical sliding guides 7.

(19) The main tilting door 1 also comprises at least two movable uprights 8 and two movable crosspieces 9 associable with each other to define a movable frame 10 that is substantially rectangular and movable with respect to the fixed frame 4. The assembly method of the movable

uprights **8** and of the movable crosspieces **9** to obtain the movable frame **10** will be described in detail later on in this treatise.

(20) In order to allow the opening and closing motion of the main tilting door **1**, tilting movement means **11** are provided, which are adapted to guide the movement of the movable frame **10** with respect to the fixed frame **4**.

(21) In particular, thanks to the tilting movement means **11** the movable frame **10** is movable between a closed position, in which it is arranged vertically, and an open position, in which it is arranged at the top of the fixed uprights **2** parallel to the road surface.

(22) The tilting movement means **11** comprise at least two lateral supporting arms **12**, each having a first end **13** which can be hinged to an intermediate area of the movable uprights **8** and a second end **14** which can be hinged to the top of the fixed uprights **2**.

(23) More in detail, the first end **13** of each lateral supporting arm **12** is hinged to a small block **15** attached to a respective movable upright **8**.

(24) The second end **14** of each lateral supporting arm **12**, on the other hand, is coupled to a hinge pin **16** placed at the top of a respective fixed upright **2**. The lateral supporting arms **12** have the function of guiding the movement of the movable frame **10** and of supporting the weight thereof.

(25) The tilting movement means **11** comprise at least two sliding supports **17** which can be associated with the movable uprights **8** and movable along the vertical sliding guides **7** defined on the fixed uprights **2**. The sliding supports **17** consist, e.g., of rollers that roll inside the vertical sliding guides **7**.

(26) In order to couple the sliding supports **17** to the vertical sliding guides **7**, each of them comprises at least one discharge portion **18**, i.e. an interruption of the vertical sliding guides **7** adapted to allow the insertion of said sliding supports **17** inside these.

(27) Each of the sliding supports **17** is attached to a respective movable upright **8** at a point comprised between the base of the movable upright **8** and the aforementioned intermediate hinging area of the lateral supporting arms **12**.

(28) To facilitate the movement of the movable frame **10** between the closed position and the open position, the top of each fixed upright **2** comprises at least one pulley assembly **19**.

(29) Each pulley assembly **19** is provided with: at least one pair of half-shells **20**, made of plastic material; and at least one pulley **21**, pivotally associated with the pair of half-shells **20** and on which a rope **22**, connected to a respective movable upright **8** and to at least one counterweight **23** can be at least partially wound.

(30) The pair of half-shells **20**, in actual facts, defines a container inside which the pulley **21** is housed.

(31) The half-shells **20** have reinforcement ribs **24** which give them high strength. The half-shells **20** assembled together, in fact, have a load-bearing function and not only do they support the pulleys **21** arranged inside them, but they also support the hinge pin **16**, to which the lateral supporting arms **12** are hinged, and join the fixed uprights **2** to the fixed crosspiece **3**; in this regard, it should be noted that the coupling portions **5** are obtained on the half-shells **20**.

(32) In the context of the present invention, moreover, the term “rope” is to be understood as any flexible element which, wound on the pulleys **21**, can perform the function of a proper rope, hence also a belt, chain or the like.

(33) The rope **22** has one end associated with a rope support **25**, which is fixed permanently to a respective movable upright **8**, and an opposite end associated with a respective counterweight **23**.

(34) The counterweights **23** are arranged below the pulleys **21** and are movable vertically along the fixed uprights **2**.

(35) Conveniently, the fixed uprights **2** are associated with the protective casing **26** inside which the counterweights **23** move.

(36) The function of the counterweight **23** is to facilitate the movement of the movable frame **10** when opening and closing the main tilting door **1**.

(37) The setting of the pulleys **21** in rotation and the movement of the relevant counterweights **23** is preferably obtained by the fact that the main tilting door **1** is provided with at least one electric motor unit **27** to allow for the automated opening and closing.

(38) The electric motor unit **27** is assembled on the fixed crosspiece **3**, e.g. inside a protective casing not shown in the figures; in practice, thanks to the protective casing, the electric motor unit **27** is incorporated inside the fixed crosspiece **3** and is not visible outside the main tilting door **1** from the inside of the space.

(39) The electric motor unit **27** comprises a motor **28** adapted to set a drive shaft **29** in rotation.

(40) The drive shaft **29** extends along the substantial entirety of the fixed crosspiece **3** up to the tops of the fixed uprights **2**.

(41) The ends of the drive shaft **29** can be associated by interlocking with the respective pivoting pins **30** mounted on the half-shells **20**.

(42) When the pivoting pins **30** are coupled by interlocking to the drive shaft **29**, they transmit motion to the pulleys **21** by means of a chain mechanism **31** which sets in rotation a geared pinion **32** meshing with a gearing **33** obtained on each pulley **21** (FIG. 2a).

(43) In turn, the pulleys **21**, by rotating, move the counterweight **23** and allow the opening and closing of the main tilting door **1**.

(44) In the closed position, the movable frame **10** is lockable to the fixed crosspiece **3** by means of a locking device **34** which prevents it from opening.

(45) In more detail, the locking device **34** comprises at least one clamping element **35**, preferably two, associated movable in rotation with the fixed crosspiece **3** and protruding therefrom to intercept the movable frame **10**.

(46) To allow the main tilting door **1** to open, the clamping element **35** is rotated, thus freeing the movable frame **10** and allowing it to be moved.

(47) The locking device **34** is conveniently associated with the electric motor unit **27** so that the locking/release of the movable frame **10** in the closed position can also be automated.

(48) The main tilting door **1** also comprises a plurality of panels **36** associable with the movable frame **10** to form an overhead door **37**.

(49) In particular, each of the aforementioned panels **36** has a substantially rectangular shape, and has two long sides opposed to each other and two short sides opposed to each other.

(50) In addition, the panels **36** comprise at least one of either a connecting portion **38** or a receiving portion **39**, complementary to each other, adapted to allow the association of each panel **36** with at least one adjacent panel **36**.

(51) In more detail, the connecting portion **38** and the receiving portion **39** are defined on at least one of the long sides of the panels **36**.

(52) The panels **36** can be fixed to the movable frame **10** as follows.

(53) At least one of either the movable uprights **8** or the movable crosspieces **9** comprises at least one sliding track **40** and a plurality of cursors **41** movable along the sliding track **40**.

(54) Advantageously, the sliding track **40** comprises at least one profiled element **42** defining a longitudinal cavity **43**, adapted to house the cursors **41**, and provided with two longitudinal folds **44** spaced apart from each other to define a longitudinal fissure **45**.

(55) The profiled element **42** extends along the entire length of at least one of either the movable uprights **8** or the movable crosspieces **9**.

(56) In the embodiment shown in the figures, the sliding tracks **40** are defined on both movable uprights **8** and both movable crosspieces **9**.

(57) It should also be noted that, again in the embodiment shown in the figures, the movable uprights **8** and the movable crosspieces **9** coincide with the sliding tracks **40** and with the profiled elements **42**.

(58) The cursors **41** conveniently comprise at least one protruding portion **46** defined on one side of the cursor **41**, adapted to guide the sliding thereof along the longitudinal fissure **45**.

(59) The panels are associable with the cursors **41**, in particular by interposition of the supporting brackets **47**.

(60) Advantageously, the cursors **41** are provided with first holes **48** adapted to allow the association of the cursors **41** with the supporting brackets **47** by means of first threaded connection elements **49** which can be screwed into the aforementioned first holes **48** and passing through the longitudinal fissure **45**. In turn, the panels **36** are provided with the second holes **50** adapted to allow the association of the panels **36** with the supporting brackets **47** by means of second threaded connection elements **51** which can be screwed into the aforementioned second holes **50**.

(61) The screwing of the first threaded connection elements **49** into the first holes **48** allows the cursors **41** and the supporting brackets **47** to be tightened in a sandwich-like manner on opposite sides of the longitudinal folds **44**, blocking the cursors **41** and the supporting brackets **47** along the sliding tracks **40**.

(62) Afterwards, the screwing of the second threaded connection elements **51** into the second holes **50** allows the panels **36** to be fixed to the supporting brackets **47** and then to the movable frame **10**.

(63) Preferably the first holes **48** and the second holes **50** are threaded and have a nut screw into which the first threaded connection elements **49** and the second threaded connection elements **51** can be screwed. Alternatively, the cursors **41** and the panels **36** can be provided with the respective holes **48**, **50** unthreaded and, in this case, the corresponding threaded connection elements **49**, **51** are of the type of self-tapping screws. The cursors **41** can be conveniently used to fix the small block **15**, the rope support **25** and/or the sliding supports **17** in the desired position of the movable uprights **8** in the same way as for the supporting brackets **47**.

(64) The particular solution of providing movable cursors **41** along the sliding tracks **40** to which the panels **36** can be fixed makes it possible to place the cursors **41** along the movable uprights **8** and the movable crosspieces **9** depending on the size of the panels **36** and on their assembly position on the movable frame **10**, considerably simplifying the assembly operations of the main tilting door **1** and its assembly on site.

(65) The particular solution of providing the profiled elements **42** defining a longitudinal cavity **43** is also particularly convenient for the assembly and mounting operations of the movable frame **10**.

(66) In this regard, it should be noted that, advantageously, the main tilting door **1** is provided with at least one “L”-shaped angular junction element **52** insertable into the aforementioned longitudinal cavities **43** of one of the movable uprights **8** and of one of the adjacent movable crosspieces **9**.

(67) After the angular junction elements **52** have been coupled by interlocking to the sliding tracks **40**, the movable uprights **8** and the movable crosspieces **9** are locked in position by means of screws and/or bolts.

(68) This allows the movable uprights **8** and the movable crosspieces **9** to be joined together and allows the movable frame **10** to be assembled quickly and easily, even during assembly on site.

(69) The cursors **41**, the angular junction elements **52** and the supporting brackets **47** can be made of plastic material, which gives them a reduced weight compared to the use of metal materials, while maintaining a high resistance to mechanical stress.

(70) The present invention also relates to a method for the assembly on site of a main tilting door **1**.

(71) This method comprises the transport phase of an assembly kit of a main tilting door **1** at the point where the access opening of the room is located, in which the assembly kit comprises: at least the two fixed uprights **2** and the fixed crosspiece **3**, wherein the fixed uprights **2** comprise the vertical sliding guides **7**; at least the two movable uprights **8** and the two movable crosspieces **9**; at least the two lateral supporting arms **12**; at least the two sliding supports **17**; the plurality of panels **36**.

(72) In particular, during transport, the fixed uprights **2** and the fixed crosspiece **3** are disassembled from each other, and the movable uprights **8**, the movable crosspieces **9** and the plurality of panels **36** are disassembled from each other.

(73) In the aforementioned assembly kit, the lateral supporting arms **12** are pre-assembled to the

movable uprights **8** and, in particular, the first end **13** of the lateral supporting arms **12** is hinged at the point where the intermediate area of the movable uprights **8** is located, while the second end **14**, opposed to the first end **13**, is free.

(74) In addition, in the aforementioned assembly kit the movable uprights **8** are supplied with the rope supports **25** pre-assembled to the movable uprights **8** and the ropes **22** already associated with the rope supports **25**.

(75) Conveniently, the assembly kit also comprises: the sliding tracks **40** associated with the movable uprights **8** and with the movable crosspieces **9**; the plurality of cursors **41** movable along the sliding tracks **40**; the supporting brackets **47** interposable between the cursors **41** and the panels **36**; the “L”-shaped angular junction elements **52** adapted to join the movable uprights **8** and the movable crosspieces **9**.

(76) According to the invention, after the assembly kit has been transported in the proximity of the access opening of the space, the fixed uprights **2** and the fixed crosspiece **3** are assembled together to obtain the fixed frame **4**.

(77) In more detail, the fixed crosspiece **3** is interposed perpendicularly with respect to the fixed uprights **2** and joined to them by interlocking through the prismatic coupling between the coupling portions **5** and the coupling seats **6**.

(78) After the coupling portions **5** have been coupled by interlocking to the coupling seats **6**, the fixed uprights **2** and the fixed crosspiece **3** are locked in position using screws and/or bolts.

(79) During the assembly of the fixed frame **4**, the drive shaft **29** is coupled by interlocking to the pivoting pins **30** connected to the pulleys **21**.

(80) The fixed frame **4** is then fixed to the access opening of the space, where the fixed crosspiece **3** is fixed to the lintel A of the access opening and each of the fixed uprights **2** is fixed to a respective shoulder S of the access opening.

(81) Afterwards, the movable uprights **8** and the movable crosspieces **9** are assembled with each other to obtain the substantially rectangular movable frame **10**.

(82) In particular, the assembly of the movable uprights **8** with the movable crosspieces **9** is carried out by partial insertion of the angular junction elements **52** into the sliding tracks **40**.

(83) After the angular junction elements **52** have been coupled by interlocking to the sliding tracks **40**, the movable uprights **8** and the movable crosspieces **9** are locked in position by means of screws and/or bolts.

(84) The “L” shape of the angular junction elements **52** makes it easy to join the movable uprights **8** to the movable crosspieces **9**, perpendicularly to each other.

(85) The movable frame **10** is then placed in a substantially vertical position and connected to the fixed frame **4**.

(86) In particular, the second end **14** of the lateral supporting arms **12** is hinged to the top of the fixed uprights **2** so that the weight of the movable frame **10** is supported by the lateral supporting arms **12**.

(87) Afterwards, the sliding supports **17** are engaged sliding in the vertical sliding guides **7** defined on the fixed uprights **2**.

(88) More in detail, the sliding supports **17** are inserted inside the discharge portions **18** defined in the aforementioned vertical sliding guides **7**.

(89) The sliding supports **17** are then moved along the vertical sliding guides **7** and fixed to a cursor **41** inside the sliding tracks **40**, locking them both to the corresponding movable upright **8**.

(90) At this point, the movable frame **10** is connected to the fixed frame **4** and movable with respect thereto, with a tilting movement, and the ropes **22** are wound around the pulleys **21** and associated with the counterweights **23**.

(91) Finally, the panels **36** are associated with the movable frame **10** to define the overhead door **37** and, in particular, the assembly takes place with the movable frame **10** placed in a substantially vertical position.

- (92) More in detail, one of the aforementioned panels **36** is positioned at the base of the movable frame **10**, resting on the movable crosspiece **9** positioned below, after which the cursors **41** are suitably displaced and positioned along the sliding tracks **40** so as to place them in the proximity of the second holes **50** on the panel **36**.
- (93) The panel **36** is then fixed to the movable frame **10** by interposition of the supporting brackets **47** which in turn are fixed to the cursors **41** and to the panel **36** itself.
- (94) Another of the aforementioned panels **36** is then mounted by interlocking on top of the previous one, and the previous phases of displacing the cursors **41** and fixing the supporting brackets **47** are repeated to allow the fixing of the aforementioned other panel **36** to the movable frame **10**.
- (95) The assembly phase of another of the panels **36** is repeated until it reaches the top of the movable frame **10**.
- (96) The supporting brackets **47** mounted on the movable uprights **8** are conveniently interposed between one panel **36** and the panel **36** adjacent thereto, in order to ensure a better fixing to each other.
- (97) It has in practice been ascertained that the described invention achieves the intended objects and in particular it is underlined that the main tilting door **1** allows simple assembly, thus ensuring practical assembly on site.
- (98) In addition, the main tilting door **1** consists of components of reduced size and weight, which allow easy transport and movement.
- (99) The components of the main door to be assembled are in fact easy to move and, in addition, have different parts made of plastic, which is a considerably lighter material than the use of metal materials.
- (100) In this regard it is important to point out that the main tilting door **1** according to the invention can be assembled on site by a small number of fitters, not necessarily highly qualified personnel, and without cumbersome and expensive lifting equipment.

Claims

1. A main tilting door for closing an access opening of a room, the main tilting door comprising: at least two fixed uprights and one fixed crosspiece associable with each other to define a fixed frame which can be fixed to one or more structures that form at least part of the access opening of the room, wherein said fixed uprights comprise vertical sliding guides; at least two movable uprights and two movable crosspieces associable with each other to define a substantially rectangular movable frame; a tilting movement apparatus of said movable frame comprising: at least two lateral supporting arms each having a respective first end which can be hinged to an intermediate area of a respective one of said movable uprights and a respective second end which can be hinged to a top of a respective one of said fixed uprights; and at least two sliding supports associable with said movable uprights and movable along said vertical sliding guides; and a plurality of panels associable with said movable frame; wherein at least one of said movable uprights and said movable crosspieces comprises at least one sliding track and a plurality of cursors movable along said sliding track, said panels being associable with said cursors, wherein said sliding track comprises at least one profiled element defining a longitudinal cavity, adapted to house said cursors, and provided with two longitudinal folds spaced apart from each other to define a longitudinal fissure, wherein said longitudinal cavity has a rectangular shaped first cross-section with two long sides and two short sides, and each of said cursors has a rectangular shaped second cross-section with two long sides and two short sides that substantially fills the first cross-section of the longitudinal cavity of said sliding track, and wherein each of said cursors comprises at least one protruding portion defined on one of said short sides of the cursor, said protruding portions adapted to guide the sliding of the cursors along said longitudinal fissure.

2. The main tilting door according to claim 1, wherein said panels are associable with said cursors by interposition of supporting brackets.
 3. The main tilting door according to claim 2, wherein said cursors are provided with first holes adapted to allow an association of said cursors with said supporting brackets via first threaded connection elements which can be screwed into said first holes and pass through said longitudinal fissure, the screwing of said first threaded connection elements into said first holes being adapted to tighten in a sandwich-like manner said cursors and said supporting brackets on opposite sides of said longitudinal folds to lock said cursors and said supporting brackets relative to said sliding track.
 4. The main tilting door according to claim 3, wherein said panels are provided with second holes adapted to allow an association of said panels with said supporting brackets via second threaded connection elements which can be screwed into said second holes.
 5. The main tilting door according to claim 1, wherein said profiled element extends along the entire length of the at least one of said movable uprights and said movable crosspieces.
 6. The main tilting door according to claim 1, wherein each of said panels comprises at least one of a connecting portion and a receiving portion, complementary to each other, adapted to allow an association of one of said panels with at least one adjacent panel of said plurality of panels.
 7. The main tilting door according to claim 6, wherein each of said panels has a substantially rectangular shape and has two long sides opposed to each other and two short sides opposed to each other, at least one of said long sides of said one of said panels defining said connecting portion and at least one of said long sides of said at least one adjacent panel defining said receiving portion.
 8. The main tilting door according to claim 1, further comprising: at least one "L"-shaped angular junction element insertable into a first longitudinal cavity of one of said movable uprights and into a second longitudinal cavity of one of said movable crosspieces, wherein said one of said movable uprights and said one of said movable crosspieces are adjacent to each other.
 9. The main tilting door according to claim 1, wherein said vertical sliding guides comprise discharge portions adapted to allow the insertion of said sliding supports in said vertical sliding guides.
 10. The main tilting door according to claim 1, wherein each of said fixed uprights comprises at least one coupling portion which can be coupled to a respective coupling seat arranged on said fixed crosspiece.
 11. The main tilting door according to claim 1, wherein said top of each of said fixed uprights comprises at least one pulley assembly provided with at least one pair of half-shells, made of plastic material, and with at least one pulley, pivotally associated with said pair of half-shells, and on which a rope, connected to a respective one of said movable uprights and to at least one counterweight, can be at least partially wound.
 12. The main tilting door according to claim 11, further comprising: at least one electric motor unit assembled on said fixed crosspiece and adapted to set said pulleys in rotation.
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